

SMART MULTI-SPECIALITY HOSPITAL NETWORK

Complete VLAN Information Table — All Sites

Site / Router	VLAN ID	Name	Port Range	Block Size	Network Address	Prefix	Default Gateway	Usable Hosts
Main Hospital HQ (R2)	10	DMZ	Fa0/1-5	16	10.0.1.160	/28	10.0.1.161	14
Main Hospital HQ (R2)	20	DOCTORS_ICU	Fa0/6-10	64	10.0.1.64	/26	10.0.1.65	62
Main Hospital HQ (R2)	30	BILLING_ADMIN	Fa0/11-15	64	10.0.1.0	/26	10.0.1.1	62
Main Hospital HQ (R2)	50	GUEST_WIFI	Fa0/16	256	10.0.0.0	/24	10.0.0.1	254
Diagnostic Center (R5)	100	LAB	Fa0/1-10	64	10.1.0.0	/26	10.1.0.1	62
Diagnostic Center (R5)	110	ADMIN	Fa0/11-15	32	10.1.0.64	/27	10.1.0.65	30
Diagnostic Center (R5)	120	STAFF	Fa0/16-20	16	10.1.0.96	/28	10.1.0.97	14
Emergency Clinic (R7)	200	EMERGENCY	Fa0/1-10	64	10.2.0.0	/26	10.2.0.1	62
Emergency Clinic (R7)	210	ADMIN	Fa0/11-15	32	10.2.0.64	/27	10.2.0.65	30
Emergency Clinic (R7)	220	IP_PHONE	Fa0/16-20	16	10.2.0.96	/28	10.2.0.97	14
Telemedicine (R10)	300	CONSULTATION	Fa0/1-10	64	10.3.0.0	/26	10.3.0.1	62
Telemedicine (R10)	310	ADMIN	Fa0/11-15	32	10.3.0.64	/27	10.3.0.65	30
Telemedicine (R10)	320	WIFI	Fa0/16	32	10.3.0.128	/27	10.3.0.129	30
Maternity Hospital (R12)	400	MATERNITY	Fa0/1-10	64	10.4.0.0	/26	10.4.0.1	62
Maternity Hospital (R12)	410	ADMIN	Fa0/11-15	32	10.4.0.64	/27	10.4.0.65	30
Maternity Hospital (R12)	420	WIFI	Fa0/16	32	10.4.0.128	/27	10.4.0.129	30
Children Care Hospital (R14)	500	CHILDREN	Fa0/1-10	64	10.5.0.0	/26	10.5.0.1	62
Children Care Hospital (R14)	510	ADMIN	Fa0/11-15	32	10.5.0.64	/27	10.5.0.65	30
Children Care Hospital (R14)	520	WIFI	Fa0/16	32	10.5.0.128	/27	10.5.0.129	30

VLAN CONFIGURATION — CLI CODE (ALL SITES)

1. Main Hospital HQ (R2) — Reconstructed from Packet Tracer screenshot

Note: This CLI was not in your original pasted code — reconstructed from your topology screenshot (VLAN names, port ranges, gateways). Please verify against the actual running-config on your HQ router/switch before relying on it.

Switch Configuration

```
en
conf t

vlan 10
name DMZ

vlan 20
name DOCTORS_ICU

vlan 30
name BILLING_ADMIN

vlan 50
name GUEST_WIFI

exit

int range fa0/1-5
switchport mode access
switchport access vlan 10
exit

int range fa0/6-10
switchport mode access
switchport access vlan 20
exit

int range fa0/11-15
switchport mode access
switchport access vlan 30
exit

int fa0/16
switchport mode access
switchport access vlan 50
exit

int g0/1
```

```
switchport mode trunk
switchport trunk allowed vlan 10,20,30,50
no shut
exit
```

Router Configuration

```
en
conf t
```

```
int g0/1
no shut
exit
```

```
int g0/1.10
encapsulation dot1Q 10
ip add 10.0.1.161 255.255.255.240
exit
```

```
int g0/1.20
encapsulation dot1Q 20
ip add 10.0.1.65 255.255.255.192
exit
```

```
int g0/1.30
encapsulation dot1Q 30
ip add 10.0.1.1 255.255.255.192
exit
```

```
int g0/1.50
encapsulation dot1Q 50
ip add 10.0.0.1 255.255.255.0
exit
```

```
ip dhcp excluded-address 10.0.1.161
```

```
ip dhcp pool DMZ
network 10.0.1.160 255.255.255.240
default-router 10.0.1.161
dns-server 10.0.1.162
exit
```

```
ip dhcp excluded-address 10.0.1.65
```

```
ip dhcp pool DOCTORS_ICU
network 10.0.1.64 255.255.255.192
default-router 10.0.1.65
dns-server 10.0.1.162
exit
```

```
ip dhcp excluded-address 10.0.1.1
```

```
ip dhcp pool BILLING_ADMIN  
network 10.0.1.0 255.255.255.192  
default-router 10.0.1.1  
dns-server 10.0.1.162  
exit
```

```
ip dhcp excluded-address 10.0.0.1
```

```
ip dhcp pool GUEST_WIFI  
network 10.0.0.0 255.255.255.0  
default-router 10.0.0.1  
dns-server 8.8.8.8  
exit
```

2. Diagnostic Center (R5)

Switch Configuration

```
en
conf t

vlan 100
name LAB

vlan 110
name ADMIN

vlan 120
name STAFF

exit

int range fa0/1-10
switchport mode access
switchport access vlan 100
exit

int range fa0/11-15
switchport mode access
switchport access vlan 110
exit

int range fa0/16-20
switchport mode access
switchport access vlan 120
exit

int g0/1
switchport mode trunk
switchport trunk allowed vlan 100,110,120
no shut
exit
```

Router Configuration

```
en
conf t

int g0/1
no shut
exit

int g0/1.100
encapsulation dot1Q 100
```

```
ip add 10.1.0.1 255.255.255.192
exit

int g0/1.110
encapsulation dot1Q 110
ip add 10.1.0.65 255.255.255.224
exit

int g0/1.120
encapsulation dot1Q 120
ip add 10.1.0.97 255.255.255.240
exit

ip dhcp excluded-address 10.1.0.1

ip dhcp pool LAB
network 10.1.0.0 255.255.255.192
default-router 10.1.0.1
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.1.0.65

ip dhcp pool ADMIN
network 10.1.0.64 255.255.255.224
default-router 10.1.0.65
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.1.0.97

ip dhcp pool STAFF
network 10.1.0.96 255.255.255.240
default-router 10.1.0.97
dns-server 10.0.1.162
exit
```

3. Emergency Clinic (R7)

Switch Configuration

```
en
conf t

vlan 200
name EMERGENCY

vlan 210
name ADMIN

vlan 220
name IP_PHONE

exit

int range fa0/1-10
switchport mode access
switchport access vlan 200
exit

int range fa0/11-15
switchport mode access
switchport access vlan 210
exit

int range fa0/16-20
switchport mode access
switchport access vlan 220
exit

int g0/1
switchport mode trunk
switchport trunk allowed vlan 200,210,220
no shut
exit
```

Router Configuration

```
en
conf t

int g0/1
no shut
exit

int g0/1.200
encapsulation dot1Q 200
```

```
ip add 10.2.0.1 255.255.255.192
exit

int g0/1.210
encapsulation dot1Q 210
ip add 10.2.0.65 255.255.255.224
exit

int g0/1.220
encapsulation dot1Q 220
ip add 10.2.0.97 255.255.255.240
exit

ip dhcp excluded-address 10.2.0.1
ip dhcp excluded-address 10.2.0.65
ip dhcp excluded-address 10.2.0.97

ip dhcp pool EMERGENCY
network 10.2.0.0 255.255.255.192
default-router 10.2.0.1
dns-server 10.0.1.162
exit

ip dhcp pool EMERGENCY_ADMIN
network 10.2.0.64 255.255.255.224
default-router 10.2.0.65
dns-server 10.0.1.162
exit

ip dhcp pool IP_PHONE
network 10.2.0.96 255.255.255.240
default-router 10.2.0.97
dns-server 10.0.1.162
exit
```


4. Telemedicine Branch (R9/R10)

Switch Configuration

```
en
conf t

vlan 300
name CONSULTATION

vlan 310
name ADMIN

vlan 320
name WIFI

exit

int range fa0/1-10
switchport mode access
switchport access vlan 300
exit

int range fa0/11-15
switchport mode access
switchport access vlan 310
exit

int fa0/16
switchport mode access
switchport access vlan 320
exit

int g0/1
switchport mode trunk
switchport trunk allowed vlan 300,310,320
no shut
exit
```

Router Configuration

```
en
conf t

int g0/1
no shut
exit

int g0/1.300
encapsulation dot1Q 300
```

```
ip add 10.3.0.1 255.255.255.192
exit

int g0/1.310
encapsulation dot1Q 310
ip add 10.3.0.65 255.255.255.224
exit

int g0/1.320
encapsulation dot1Q 320
ip add 10.3.0.129 255.255.255.224
exit

ip dhcp excluded-address 10.3.0.1

ip dhcp pool CONSULTATION
network 10.3.0.0 255.255.255.192
default-router 10.3.0.1
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.3.0.65

ip dhcp pool TELE_ADMIN
network 10.3.0.64 255.255.255.224
default-router 10.3.0.65
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.3.0.129

ip dhcp pool WIFI
network 10.3.0.128 255.255.255.224
default-router 10.3.0.129
dns-server 10.0.1.162
exit
```

5. Maternity Hospital (R11/R12) — WIFI VLAN 420 Added

Switch Configuration

```
en
conf t

vlan 400
name MATERNITY

vlan 410
name ADMIN

vlan 420
name WIFI

exit

int range fa0/1-10
switchport mode access
switchport access vlan 400
exit

int range fa0/11-15
switchport mode access
switchport access vlan 410
exit

int fa0/16
switchport mode access
switchport access vlan 420
exit

int g0/1
switchport mode trunk
switchport trunk allowed vlan 400,410,420
no shut
exit
```

Router Configuration

```
en
conf t

int g0/1
no shut
exit

int g0/1.400
encapsulation dot1Q 400
```

```
ip add 10.4.0.1 255.255.255.192
exit

int g0/1.410
encapsulation dot1Q 410
ip add 10.4.0.65 255.255.255.224
exit

int g0/1.420
encapsulation dot1Q 420
ip add 10.4.0.129 255.255.255.224
exit

ip dhcp excluded-address 10.4.0.1

ip dhcp pool MATERNITY
network 10.4.0.0 255.255.255.192
default-router 10.4.0.1
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.4.0.65

ip dhcp pool MAT_ADMIN
network 10.4.0.64 255.255.255.224
default-router 10.4.0.65
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.4.0.129

ip dhcp pool WIFI
network 10.4.0.128 255.255.255.224
default-router 10.4.0.129
dns-server 10.0.1.162
exit
```

6. Children Care Hospital (R13/R14)

Switch Configuration

```
en
conf t

vlan 500
name CHILDREN

vlan 510
name ADMIN

vlan 520
name WIFI

exit

int range fa0/1-10
switchport mode access
switchport access vlan 500
exit

int range fa0/11-15
switchport mode access
switchport access vlan 510
exit

int fa0/16
switchport mode access
switchport access vlan 520
exit

int g0/1
switchport mode trunk
switchport trunk allowed vlan 500,510,520
no shut
exit
```

Router Configuration

```
en
conf t

int g0/1
no shut
exit

int g0/1.500
encapsulation dot1Q 500
```

```
ip add 10.5.0.1 255.255.255.192
exit

int g0/1.510
encapsulation dot1Q 510
ip add 10.5.0.65 255.255.255.224
exit

int g0/1.520
encapsulation dot1Q 520
ip add 10.5.0.129 255.255.255.224
exit

ip dhcp excluded-address 10.5.0.1

ip dhcp pool CHILDREN
network 10.5.0.0 255.255.255.192
default-router 10.5.0.1
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.5.0.65

ip dhcp pool CHILD_ADMIN
network 10.5.0.64 255.255.255.224
default-router 10.5.0.65
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.5.0.129

ip dhcp pool WIFI
network 10.5.0.128 255.255.255.224
default-router 10.5.0.129
dns-server 10.0.1.162
exit
```